

Medetomidine in the Illicit Drug Supply

Lynch MJ, Pizon AF, Yealy DM. Ann Emerg Med. 2026;87:709-716.
Implications for emergency care and withdrawal management

Clinical question

How should clinicians recognize and manage medetomidine-associated intoxication and withdrawal when it appears with illicit opioids?

What the paper says

- Medetomidine is a veterinary alpha-2 agonist increasingly detected with illicit fentanyl.
- Naloxone may restore ventilation while medetomidine sedation persists.
- Withdrawal can begin within 4-6 hours and include severe autonomic instability, vomiting, agitation, delirium, and hyperthermia.
- The authors describe early antiemetics, oral/transdermal alpha-2 agonists, and IV dexmedetomidine for refractory cases.

Evidence guardrails

- This is a narrative review/clinical-concepts paper, not a randomized treatment trial or validated guideline.
- Medetomidine-specific evidence comes largely from case reports, case series, pharmacology, and expert experience.
- Illicit exposures are polysubstance exposures; attribution and testing are limited.
- Protocol doses require local toxicology, pharmacy, emergency, and ICU review.

Do not equate an elevated COWS score with opioid withdrawal when symptoms occur unusually soon after last use. Reassess timing, physiology, and likely co-exposures, and obtain toxicology or poison-center support.

Clinical pattern

Intoxication

- Prolonged sedation, miosis, bradycardia, and initial hypertension followed by hypotension.
- Naloxone does not reverse medetomidine; supportive care and airway protection remain central.

Withdrawal

- Anxiety, tremor, diaphoresis, vomiting, tachycardia, hypertension, agitation, delirium, and hyperthermia.
- Possible hypokalemia, acidosis, QTc prolongation, cardiomyopathy, and encephalopathy.

Proposed management framework

Recognize and monitor

- Clinical diagnosis; airway, breathing, circulation, ECG, electrolytes, mental status, and trajectory.
- Observe for delayed progression and escalate persistent toxicity or severe withdrawal.

Stabilize and treat

- Early dopamine-antagonist antiemetic therapy; oral clonidine plus scheduled guanfacine; transdermal clonidine for tapering.
- IV dexmedetomidine may be needed for refractory cases; manage OUD separately and avoid premature buprenorphine based only on COWS.

Professional-context update: CDC materials emphasize that atipamezole is not an established human reversal strategy and may precipitate dangerous withdrawal. CDC, UPMC, and PubMed-indexed education reinforce toxicology consultation and the evolving evidence base.

Discussion prompts

- 1 Which features best distinguish medetomidine withdrawal from opioid, benzodiazepine, stimulant, or xylazine withdrawal?
- 2 When is COWS insufficient for deciding buprenorphine timing?
- 3 Which parts of the proposed algorithm are feasible in our setting?
- 4 How should suspected exposure be documented when confirmatory testing is unavailable?
- 5 What harm-reduction and discharge instructions are needed as the drug supply changes?
- 6 What prospective data would most change these recommendations?

Facilitator discussion

The most useful discriminator is the time course and physiologic pattern: medetomidine withdrawal can begin unusually soon, with prominent vomiting, tremor, hypertension, tachycardia, agitation, delirium, and hyperthermia. Opioid withdrawal may overlap, while benzodiazepine withdrawal more often centers on anxiety, perceptual change, seizures, and delirium; stimulant toxicity tends toward sympathomimetic activation; xylazine can produce sedation, bradycardia, hypotension, and its own withdrawal syndrome. None is reliably diagnosed by a single score, and polysubstance exposure means the team should document a working differential rather than force one label.

COWS is insufficient when symptoms are early, atypical, refractory to expected opioid treatment, or dominated by autonomic instability. In practice, the feasible core is structured monitoring, ECG/electrolytes, antiemetic access, alpha-2 agonist protocols, clear ICU/toxicology escalation, and separate reassessment for MOUD timing. When testing is unavailable, document suspected exposure as a clinical syndrome with last known use, response to naloxone and opioid-directed therapy, vital-sign trajectory, exam findings, co-exposures, consultant input, and uncertainty. Discharge planning should include naloxone, not using alone, safer supply and sterile equipment counseling, return precautions for delayed vomiting or autonomic symptoms, and linkage to MOUD. The highest-value prospective evidence would combine a validated withdrawal scale with a treatment registry, confirmatory toxicology, and comparative protocols.

Clinical implications for ASAM 3.7/3.5 practice

Operational takeaway: Ask about changing drug-supply effects when a patient has prolonged sedation after naloxone or develops unusually early, severe, refractory autonomic withdrawal. Arrange monitoring that matches the trajectory, not just the initial presentation.

- Build medetomidine awareness into intake and handoff language without assuming the patient can identify the adulterant.
- Do not use opioid withdrawal as the only explanatory label for early severe symptoms.
- Maintain a clear escalation pathway to emergency medicine, medical toxicology, poison center, ICU, pharmacy, and addiction medicine.
- Separate acute stabilization from eventual MOUD initiation; reassess physiology, timing, opioid exposure, and withdrawal phenotype.
- Provide return precautions for delayed vomiting, diaphoresis, tremor, agitation, chest pain, or vital-sign abnormalities.

Strengths, limitations, and professional context

Strengths

- Clinically useful description of a rapidly evolving toxidrome.
- Connects medetomidine observations with dexmedetomidine withdrawal literature.
- Provides a concrete multidisciplinary treatment framework.
- Separates opioid respiratory-depression reversal from alpha-2 sedation.

Limitations

- No denominator, comparator, randomized evidence, or validated severity scale.
- Testing and attribution are uncertain in polysubstance exposures.
- Protocol doses and escalation thresholds may reflect UPMC resources.
- The evidence base is early and treatment recommendations remain provisional.

References and professional resources

Primary article: <https://doi.org/10.1016/j.annemergmed.2025.12.004>

Professional context: CDC clinician webinar and advisory; PubMed clinical update; UPMC clinical education; American Family Physician commentary.

This handout is for journal-club education and clinical discussion. Local emergency, toxicology, pharmacy, and critical-care guidance takes precedence.

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